Index Number:

Candidate Name: ______________________________ CT Group: 07S____

# HWA CHONG INSTITUTION

## JC2 PRELIMINARY EXAMINATIONS

## H2 BIOLOGY 9747

## PAPER 3

15 Sept 2008 1 hr 30 min

**INSTRUCTIONS TO CANDIDATES**

There are **3** question booklets (**I** to **III**) to this paper. Write your **name**, **CT** and **Index Number** on the cover pages of the booklets in the spaces provided. Write in blue or black pen.

**SECTION A**

This section contains **3** questions. Answer **ALL** questions.

Write your answers in the spaces provided on the question paper.

**SECTION B**

This section contains **1** question with three parts (a), (b) and (c). Answer **ALL** parts.

Your answer to **Section B** must be in continuous prose, where appropriate.

**Begin each part of the question on a separate fresh sheet of answer paper.**

A NIL return is required for any part(s) you have not attempted.

At the end of the examination, submit all question booklets and answer sheets to each part of **Section B** separately.

**INFORMATION FOR CANDIDATES**

The intended number of marks is given in brackets [ ] at the at the end of each question or part question.

| FOR EXAMINERS' USE | |
|---|---|
| 1 | / 15 |
| 2 | / 15 |
| 3 | / 15 |
| 4 | / 20 |
| TOTAL | / 65 |

BOOKLET I

## SECTION A: STRUCTURED QUESTIONS

### QUESTION 1

In a study of a rare genetic disease, DNA samples from two generations of an affected family were obtained and used for Restriction Fragment Length Polymorphism (RFLP) linkage analysis. One RFLP used is found at locus **P** on chromosome 3 and identified by **Probe P** (10 kb). The other is at locus **Q** on chromosome 17 and identified by **Probe Q** (5 kb). The various alleles of the two RFLP loci are shown in **Fig. 1.1**. The results obtained from RFLP analysis are shown in **Fig. 1.2** below (not drawn to scale).

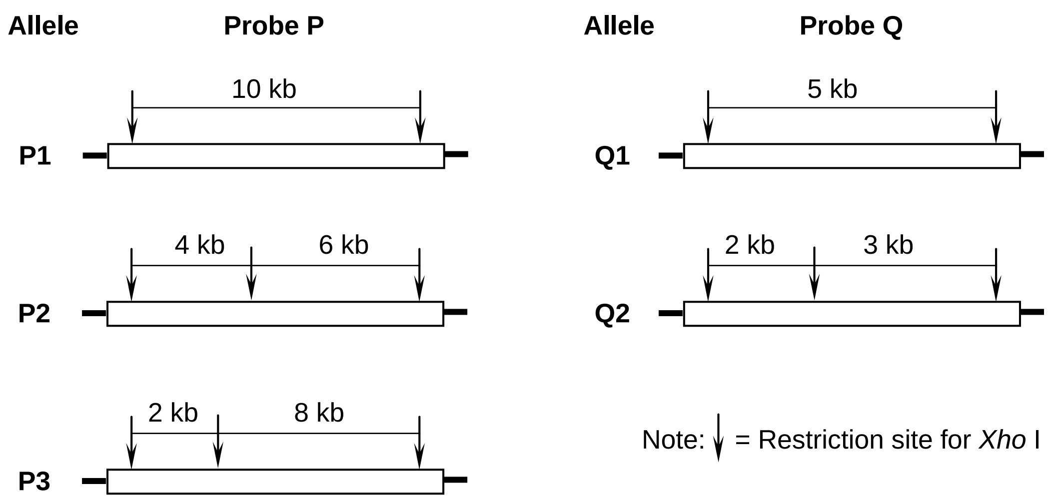


**Fig. 1.1**

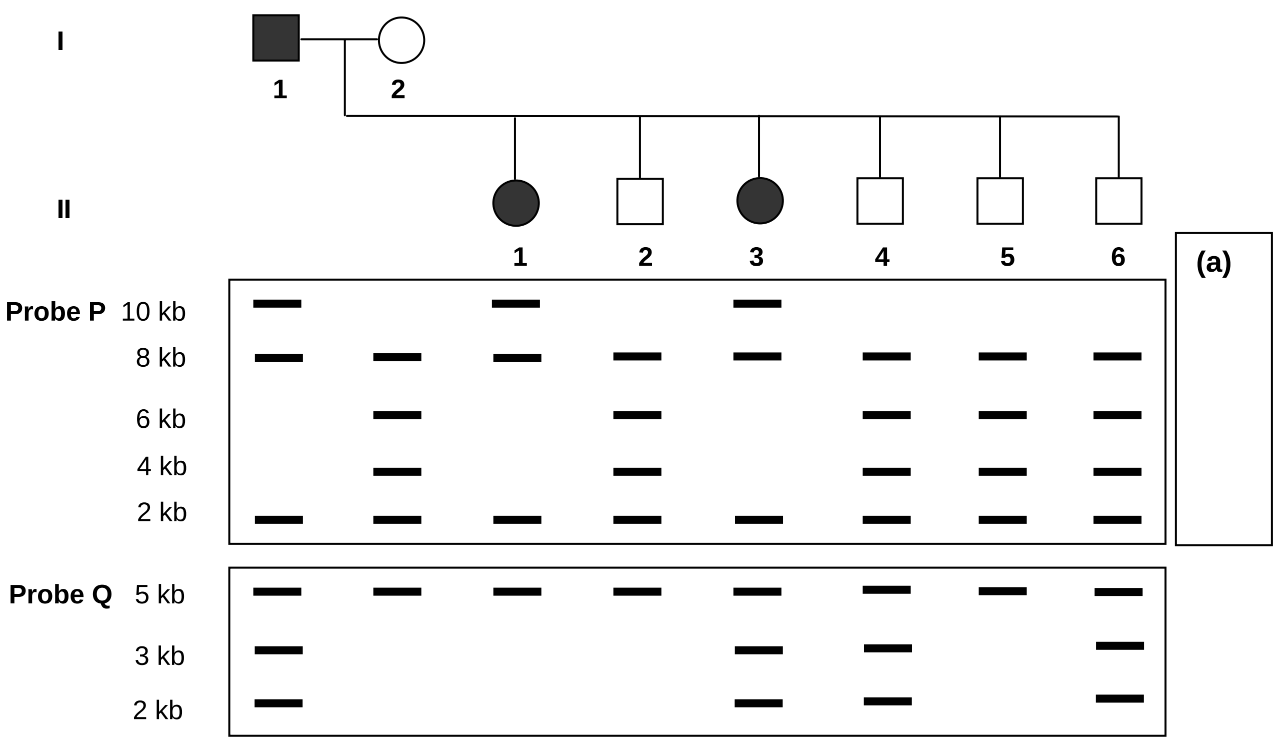


**Fig. 1.2**

**(a)** In the space provided in **Fig. 1.2**, indicate with

**(i)** the symbols ⊕ and ⊖, the relative positions of the electrodes, and

**(ii)** an **arrow,** the direction of migration of restriction fragments during gel electrophoresis.[2]

**(b)** State what is meant when a certain locus is described as 'polymorphic'. [1]

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**(c)** Describe how RFLP may be used to map the chromosomal locus of a dominant mutant allele. [3]

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**(d)** Based on the results shown in **Fig. 1.2**,

**(i)** state the genotypes of individuals **I-2** and **II-3** with respect to the **P** and **Q** loci. [2]

Individual **I-2** : ________________ ________________

Individual **II-3** : ________________ ________________

**(ii)** state on which chromosome the disease gene is located. [1]

______________________________________________________________________

**(e)** Individual **II-6** marries a woman with the genotype **P1P2** and they have a child who is homozygous for allele **P1**. The father suspects the child was illegitimate.

**(i)** Explain the genetic basis for this suspicion. [2]

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**(ii)** If the child is not illegitimate, suggest one possible explanation how the child could have acquired the **P1P1** genotype. [2]

**(iii)** Suggest, with a reason, what further test(s) may be used to confirm the suspicion. [2]

**[TOTAL: 15 marks]**

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BOOKLET II

## SECTION A: STRUCTURED QUESTIONS (Continued)

### QUESTION 2

The Salmon Genome Project (SGP) is developed to increase knowledge of the biology of Atlantic salmon and aid agricultural breeding of the fish.

**(a)** Describe how the SGP serves to increase knowledge of the biology of Atlantic salmon. [2]

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In order to produce transgenic salmon expressing salmon growth hormone, sGH, the coding sequence of sGH gene is isolated from a library and cloned, using *Sac* I, into the plasmid expression vector, pBluescript II SK. The plasmid map is shown in **Fig. 2.1**.

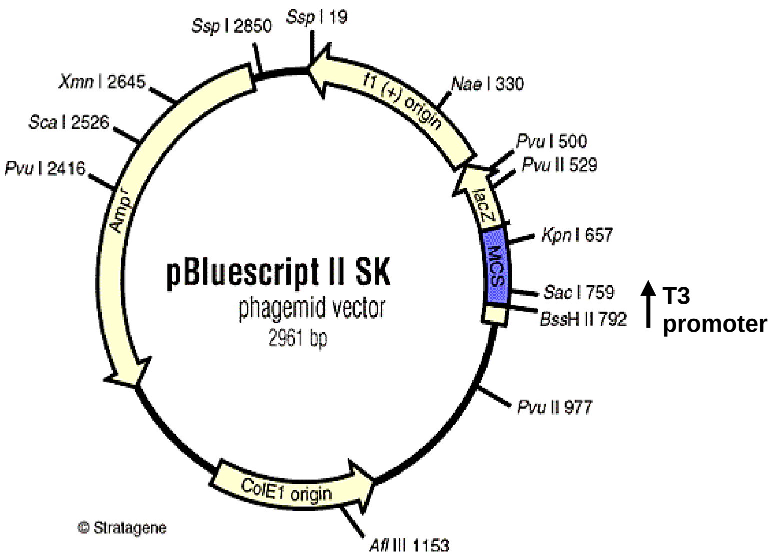


**Fig. 2.1**

**(b)** With reference to **Fig. 2.1**, describe the features of the multiple cloning site, MCS. [2]

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Subsequent to insertion of sGH gene into pBlueScript II SK, transformation into *E. coli* cells was carried out and some colonies were obtained. The plasmid DNA was extracted, digested with *Sac* I and the restriction fragments were separated in gel electrophoresis.

**(c) (i)** It was found that the recombinant plasmid with sGH gene inserted yielded no polypeptide. With reference to **Fig. 2.1**, state and explain a reason for this. [2]

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**(ii)** Suggest a solution for the problem in **(c) (i)**. [1]

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The problem was rectified and the recombinant plasmid now produces functional sGH. The recombinant plasmid was microinjected into eggs from very slow-growing domestic and wild strains of Atlantic salmon. Their growth were scored, as shown in **Fig. 2.2**. In a further analysis, growth hormone was extracted from Domestic-GH treated salmon on dates 11/5, 23/6, 14/7 and 1/9. A protein gel was run and the results shown in **Fig. 2.3**.

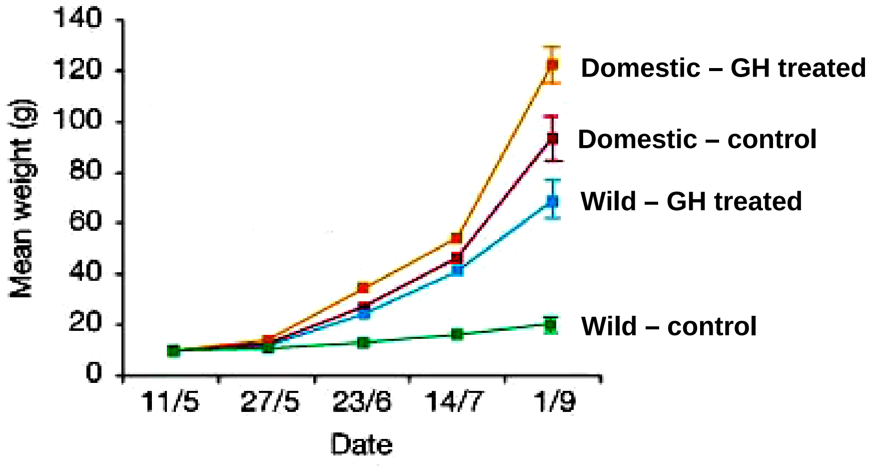



**Fig. 2.2**

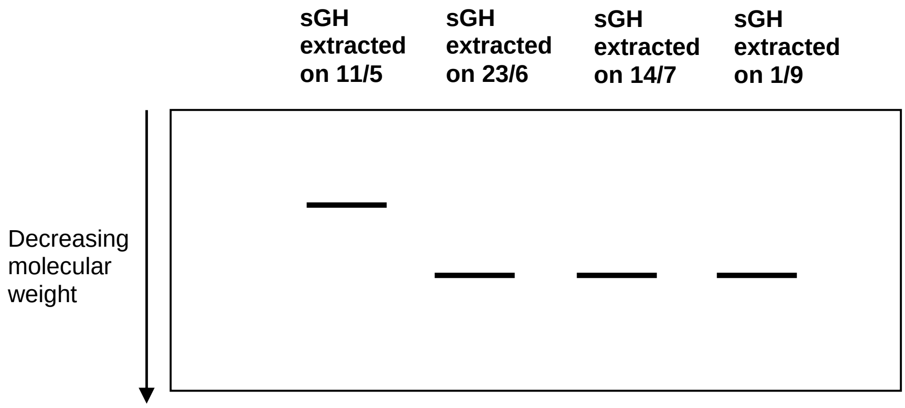



**Fig. 2.3**

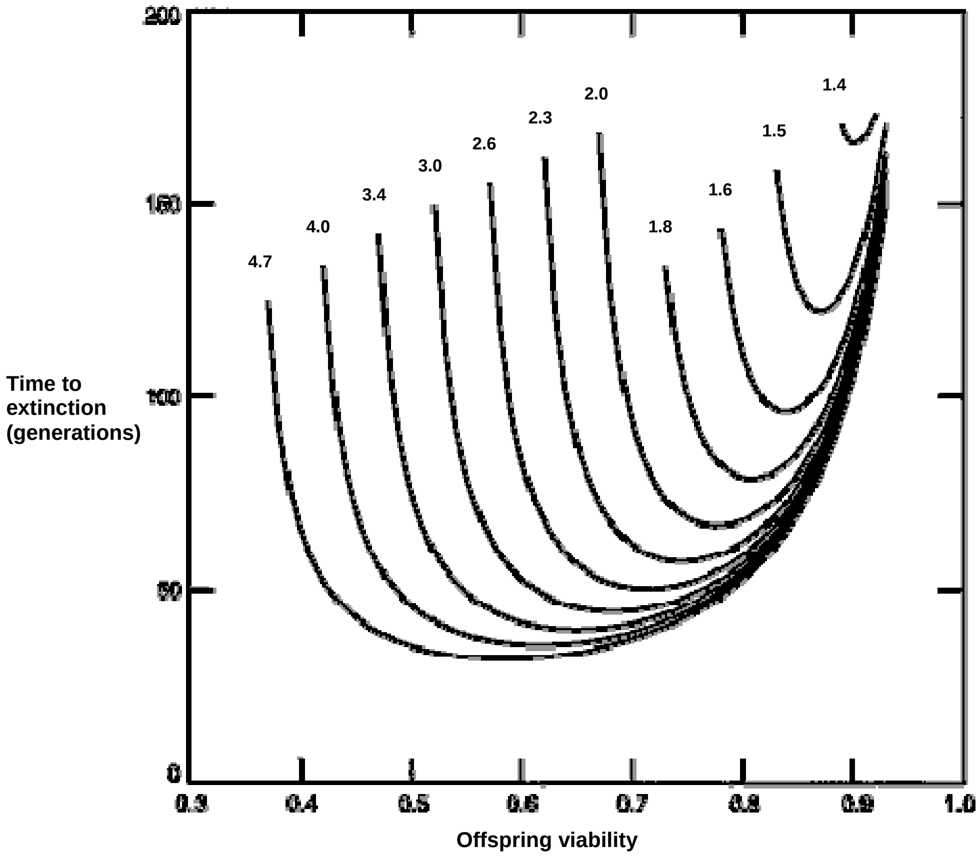
**(d)** **(i)** Describe the results shown in **Fig. 2.2.** [2]

**(ii)** Suggest a reason for the difference in weight gain between Domestic-GH treated and Wild-GH treated fish. [1]

**(iii)** Describe the results seen in **Fig. 2.3**. [1]

**(iv)** With reference to both **Fig. 2.2** and **Fig. 2.3**, suggest the process sGH undergoes before exerting its effect on weight gain. [1]

Widespread interest in producing transgenic salmon is balanced over ecological hazards, such as species extinction if such fishes were to be released into the Atlantic Ocean. A study to explore the ecological consequences of transgene release into natural populations was carried out. Two variables were considered: the ratio of the mating advantage of transgenic males relative to that of wild-type males and the relative viability of transgenic offspring. The predicted time to extinction of a wild-type salmon population was studied using these two variables. The results of the study are shown in **Fig. 2.4**.




*The numbers above the curve represent the ratio of the mating advantage of transgenic males relative to that of wild-type males.*

**Fig. 2.4**

**(e)** **(i)** With reference to **Fig. 2.4**, state the one selection pressure faced by transgenic salmon. [1]

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**(ii)** Using your answer to **(e) (i)**, explain how the named selection pressure may or may not lead to salmon extinction. [2]

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**[TOTAL: 15 marks]**

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BOOKLET III

## SECTION A: STRUCTURED QUESTIONS (Continued)

### QUESTION 3

A new brightly-coloured mouse was discovered in Russia and it was given the name *Mus colores*. **Fig. 3.1** below shows two flowcharts that illustrate how blue and green cells were isolated from different parts of this mouse, and each cell was cultured individually.

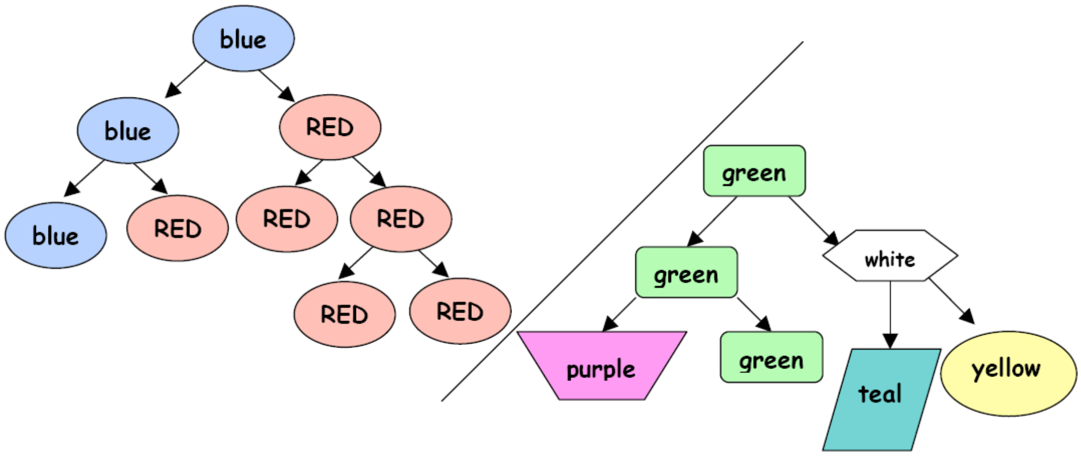


**Fig. 3.1**

With reference to **Fig. 3.1**,

**(a) (i)** name the process in which the blue cells were formed from the isolated cells. [1]

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**(ii)** are the blue cells stem cells? Give a reason for your answer. [2]

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**(iii)** given that the green cell is a stem cell, explain why green cells can give rise to differently coloured cells. [2]

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**Fig. 3.2** below shows **Steps 1 – 4** of an experimental gene therapy for insulin-dependent diabetes using mouse embryonic stem cells (mESCs). These mESCs were derived from the inner cell mass of the blastocyst of a pregnant diabetic mouse. The normal insulin gene, combined with a glucose-sensitive promoter, was inserted into the mESCs to replace the defective insulin gene. These mESCs were cultured under specific conditions. The embryonic stem cells were then expanded and differentiated. Cells with cell surface protein markers consistent with pancreatic islet cells were selected for further differentiation and characterization. When these cells were grown in culture, they spontaneously formed three-dimensional clusters similar in structure to normal pancreatic islets.

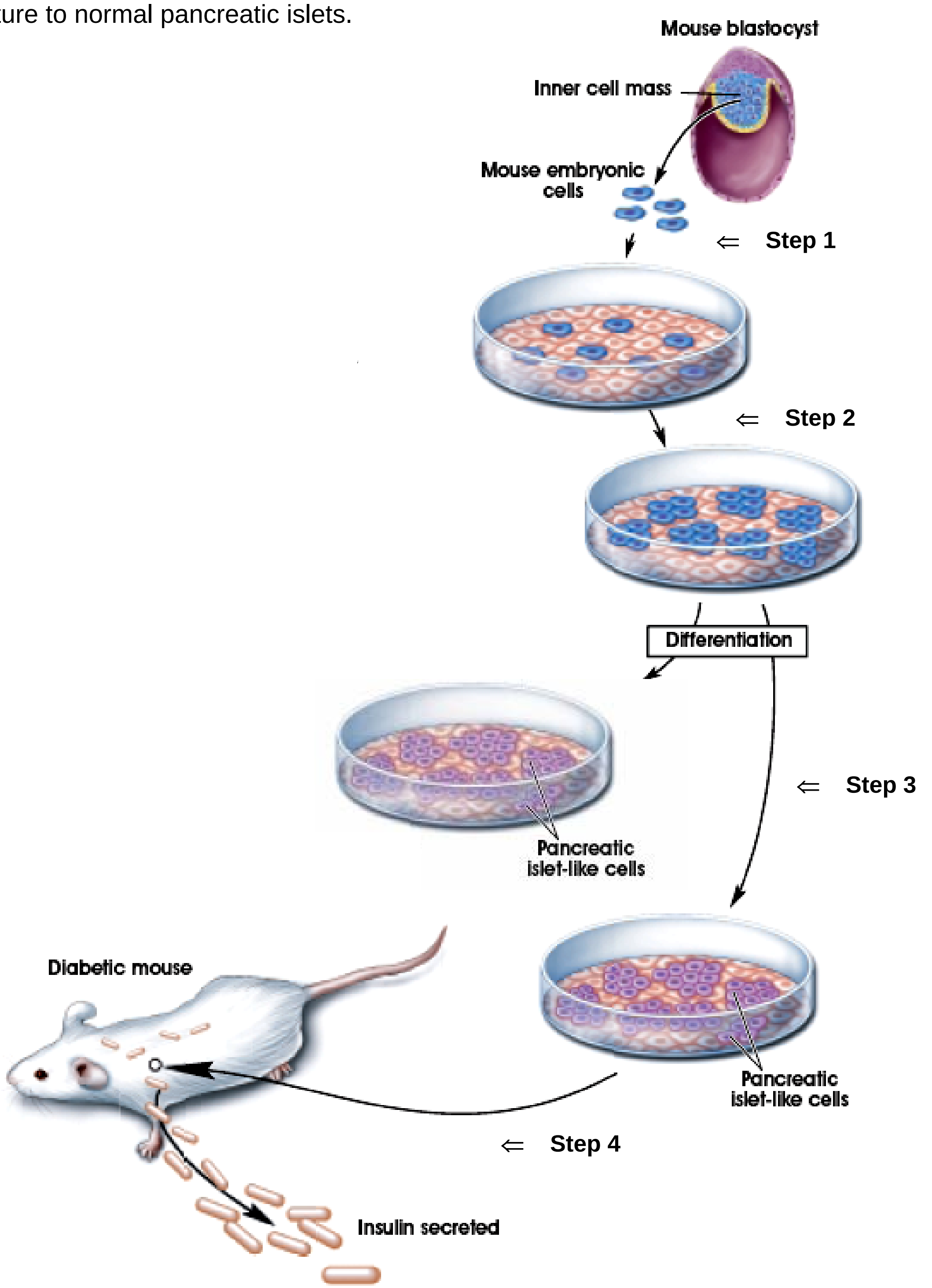


**Fig. 3.2**

With reference to **Fig. 3.2**,

**(b)** **(i)** name the type of gene therapy involved. [1]

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**(ii)** suggest why the cells in **Steps 1** to **3** were grown on other cells when cultured. [1]

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**(iii)** give a reason why cells with markers consistent with pancreatic islet cells from **Step 2** were selected for further differentiation in **Step 3**. [1]

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When the pancreatic islet-like cells from **Step 3** of **Fig. 3.2** were implanted in the shoulder of the diabetic mouse, the cells became vascularized through formation of blood vessels, synthesized insulin, and maintained physical characteristics similar to pancreatic islets. Following transplantation of the pancreatic islet-like cells, the blood glucose concentration of the mouse was taken 1 hour after every meal each day. The daily mean obtained is shown in **Fig. 3.3** below. The set point for normal blood glucose concentration is 7 mg/dm$^3$.

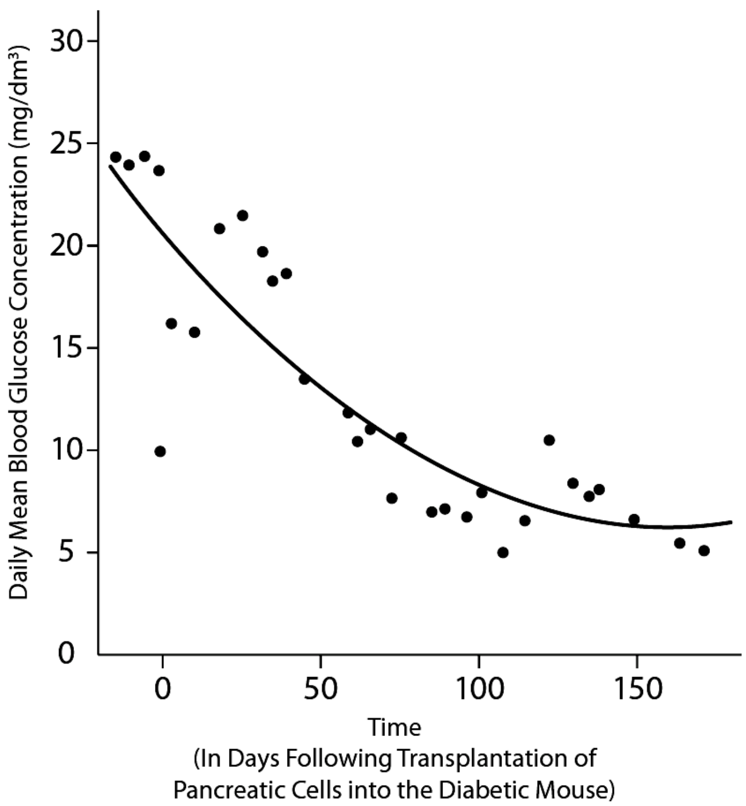


**Fig. 3.3**

**(c)** **(i)** When the pancreatic-like cells were placed in the diabetic mouse, explain briefly why it was necessary for vascularization to take place. [2]

**(ii)** Outline the role of the glucose-sensitive promoter in this method of gene therapy for mice. [1]

**(iii)** With reference to **Fig. 3.3**, account for the results obtained. [2]

**(iv)** Discuss one possible **benefit** and **problem** of using this gene therapy in the treatment of diabetes in humans. [2]

Benefit:

Problem:

**[Total: 15 marks]**

**--- End of Section A ---**

## SECTION B: FREE RESPONSE QUESTION

Your answer to this question must be in continuous prose, where appropriate.

Your answer should be illustrated by large, clearly labeled diagrams, where appropriate.

Begin each part on a fresh sheet of answer paper provided.

**QUESTION 4**

**(a)** Using a tomato leaf, describe in detail how protoplast culture can be used for the micropropagation of the plant. [8]

**(b)** Explain why farmers still use traditional methods to clone their crops despite the availability of tissue culture technologies. [4]

**(c)** Like plant transgenesis, the human genome project is a landmark scientific breakthrough. Discuss the benefits and associated ethical implications of both breakthroughs. [8]

**--- End of Section B ---**